

GeoTracker

AI Visibility Audit Platform — Technical Build Plan

Project	GeoTracker
Parent Company	lozera
Derived From	GeoTracker BRD v1.0 + SOW v1.0
Aligned Against	SOW_ALIGNMENT_v1.0 audit
Codebase Snapshot	branch claude/plan-geotracker-app-uLLsn @ 01c28d8
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Author	Allan
Approver	Timothy Ngo (Tim), CEO
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Status	Aligned to current build · Awaiting Tim sign-off

v1.1 is a corrective revision. SOW v1.0 was partly inferential; this version is aligned to a full codebase audit committed as SOW_ALIGNMENT_v1.0.md. Material corrections (auth mechanism, geospatial fallback, hosting model, data model shape, bot definitions, testing scope) are documented in §0 — Changelog. The build plan, milestones, and exit criteria are restructured to reflect what was actually built and what is genuinely remaining.

0. Changelog (v1.0 → v1.1)

The following corrections were applied based on the SOW_ALIGNMENT_v1.0 audit. Each item reflects a factual delta between SOW v1.0 and the actual codebase as of 2026-05-23. Original SOW v1.0 wording was inferential; this revision is evidence-based.

§	Change	Reason
§3.2	Auth: removed 'JWT cookies', replaced with 'HMAC-signed session cookies (custom)'	No JWT dependency exists. Custom HMAC implementation in apps/api/src/auth/session.ts
§3.2	Geospatial: removed 'Mapbox as fallback', replaced with 'curated 5-city Texas fixture'	Mapbox never integrated. MAPBOX_API_KEY env-stubbed but unused.
§3.2	Web hosting: changed 'static + edge functions' to 'SSR via @astrojs/vercel adapter'	apps/web/astro.config.mjs ships output: 'server'
§3.2	Frontend: removed 'shadcn/ui' from confirmed stack, moved to open question	Tailwind hand-styled. shadcn never pulled.
§3.4	Renamed `industry_radius` → `industries` + separate `prompt_sets` table (FK reversed)	Actual schema in apps/api/src/db/schema.ts uses this shape
§3.4	Rewrote `pulse_subscriptions` columns to (frequency, next_run_at, last_audit_id)	v1.0 columns (active, last_sent_at) never existed
§3.4	Removed `events` table from data model	Never built. Moved to Phase 1.5 backlog.
§3.4	Added `api_health` table to data model	Exists in schema; SOW v1.0 omitted it
§3.4	Documented extra columns on audits, audit_results, businesses, users	Reflect actual schema
§3.3	Updated module list: 4 LLM adapter files cover 5 providers (OpenAI-compatible shared)	ChatGPT, Perplexity, Grok share apps/api/src/llm/openaiCompatible.ts
§3.3	Added realtime/progress.ts and routes/health.ts, routes/me.ts to module list	Exist in code; SOW v1.0 omitted them
§4	Restructured milestones to align with PLAN.md's actual 10 + H1/H2 model	PLAN.md is the source of truth for ordering and granularity
§6.3	Redefined API Health Bot as passive table populated during audits (no probe, no alerts)	True shape per apps/api/src/llm/health.ts
§6.3	Moved Lead Routing Bot to Phase 1.5 backlog	Not built. /admin/leads sorts by score, no bot exists.

§	Change	Reason
§7	Moved Load + Chaos testing to Phase 1.5 backlog	Not built. Not required for Phase 1 close-out.
§5	Tightened AC-7 to require boot-time guard (not runtime route check)	Audit flagged the gap
§5	Added explicit gating language to AC-10 PostHog requirement	Audit flagged zero events firing
§8	Added 'gate preview behind basic auth' as a hard pre-deploy requirement	Mock data on public preview is a sales-trust risk
§9	Added 'Mapbox env-stub' as a backlog cleanup risk	Promised dependency that doesn't exist in code is a documentation lie

1. Purpose & Scope

Build and deploy **GeoTracker**, an AI visibility audit web application. Users paste a business domain and instantly see whether it is recommended by ChatGPT, Perplexity, Claude, Gemini, and Grok for buying-intent prompts across their service catchment area. The app functions as the primary lead generation channel for Upserv.ai and as a freemium SaaS product in its own right.

1.1 In Scope (Phase 1)

- Web application: hero, audit engine, reveal dashboard, vendor comparison, CTA flow.
- Async parallel querying of 5 LLM APIs (ChatGPT, Perplexity, Claude, Gemini, Grok) — *real-LLM mode gated behind LLM_USE_REAL_ADAPTERS flag*.
- Tiered Green / Yellow / Red scoring with contact-info downgrade rule.
- Geospatial catchment expansion using Google Maps (live path) with curated 5-city Texas fixture as offline fallback.
- Business-type radius lookup (20 industries seeded).
- OAuth (Google + LinkedIn) — code wired; OAuth credentials pending provisioning.
- Free tier complete; **premium tier scaffolded in schema only** (no paywall, no Stripe — promoted to Phase 1.5).
- Monthly Pulse Report cron + Resend email — wired; live-send pending domain verification.
- Admin dashboard: lead tracking complete; user management, API health UI, conversion funnel UI deferred to Phase 1.5.
- Real-time progress bar via SSE during audit execution.
- Dental vertical at full prompt depth; 13 generic verticals on a shared 4-prompt fallback template.

1.2 Out of Scope (Phase 2+)

- Competitive Conquest mode (user adds competitors for head-to-head audits).
- Predictive Visibility analytics (training-trend signals).
- White-label / agency reseller portal.
- Public API for power users and integrators.
- LLM coverage beyond the Phase-1 five.
- Native mobile applications.
- Verticalized prompt depth for the 13 generic industries.

1.3 Phase 1.5 Backlog (Pre-launch polish)

Items not required for code-complete status but expected before customer-facing launch. Tracked separately from Phase 2 because they may block production sign-off.

- Stripe-backed premium tier paywall and billing.
- PostHog client + server event capture (AC-10).
- Admin dashboard: user management UI, API health UI, conversion funnel UI.
- Active API Health probe (BullMQ repeatable job; outage alerting).
- Lead Routing Bot (notification on low-scoring audits).
- Internal events table for analytics mirror (if PostHog alone is insufficient).
- Load testing (k6 / Artillery) and chaos testing.
- Safari + Firefox manual QA pass.
- Vertical prompt depth for non-dental industries (content task).

1.4 Explicitly Excluded

- Upserv.ai delivery (ongoing red-X remediation) — separate workstream owned by Upserv team.
- Kriss.ai chatbot integration into GeoTracker — handled in a future cross-product epic.
- Sales rep CRM tooling — reps use existing IoZera workflow.

2. Assumptions, Constraints, Dependencies

2.1 Current State Snapshot

As of audit date 2026-05-23: code-complete on PLAN.md milestones 1–10, plus two hardening passes. Running locally end-to-end against mock LLM adapters. **Not yet deployed.** 13 of 22 environment variables are placeholders.

2.2 Assumptions

- **Pending:** API access for all 5 LLMs at developer tier with reasonable rate limits.
- **Pending:** Google Maps Platform billing enabled for Geocoding + Places API.
- **Pending:** OAuth credentials provisioned for Google and LinkedIn.
- **Pending:** Resend account with verified sending domain.
- Dental is the launch vertical; other industries follow post-launch.
- Sales rep distribution channel exists (chit-chat outreach per BRD §10).

2.3 Constraints

- Total audit latency must remain under 15 seconds P95 (BRD §7.2). *Met against mock (~340ms); unmeasured against real LLMs.*
- Phase 1 must ship to a hosted environment, not localhost-only. *Currently violated — no API host provisioned.*
- All LLM API keys, OAuth secrets, and email credentials must live in secret storage — never client-side.
- Concurrency target of 1,000+ reps deferred to load testing in Phase 1.5.

2.4 External Dependencies

Dependency	Owner	State	Required By
OpenAI API key	Tim	FAIL Placeholder	M9 (real-LLM flip)
Perplexity API key	Tim	FAIL Placeholder	M9
Anthropic API key	Tim	FAIL Placeholder	M9
Google Gemini API key	Tim	FAIL Placeholder	M9
xAI (Grok) API key	Tim	FAIL Placeholder	M9
Google Maps API key	Tim	FAIL Placeholder	Live catchment path
Google + LinkedIn OAuth creds	Tim	FAIL Placeholder	Auth go-live
Resend account + verified domain	Allan	FAIL Placeholder	Pulse Report go-live
Railway / Fly.io / Render account	Allan	FAIL Not chosen	Deploy milestone
Vercel project	Allan	PASS Adapter wired	Frontend deploy
Token budget approval	Tim	FAIL Pending (BRD §14)	M9 (real-LLM flip)

Dependency	Owner	State	Required By
Tim sign-off on SOW v1.1	Tim	FAIL Pending	Close-out

3. Technical Architecture

3.1 High-Level Topology

GeoTracker is a two-tier system: a server-rendered Astro 5 + React frontend on Vercel (via @astrojs/vercel adapter, SSR mode), and a long-running Fastify API orchestrating LLM calls asynchronously via BullMQ on Redis, persisting audits to Postgres, and streaming real-time progress to the client via Server-Sent Events through a Redis pub/sub bridge.

API host is unprovisioned as of audit. Railway is the proposed default (single project: managed Postgres + Redis + Fastify service). Fly.io and Render remain options. Decision required from Allan before deploy milestone. See §11 Open Questions.

3.2 Stack — Confirmed (As Built)

Layer	Technology	Status
Frontend	Astro 5 + React + Tailwind (hand-styled, no shadcn)	PASS Built
Backend API	Fastify (Node.js)	PASS Built
Job queue	BullMQ + Redis	PASS Built
Database	Postgres (Drizzle ORM, schema in apps/api/src/db/schema.ts)	PASS Built
Cache	Redis	PASS Built
Real-time	Server-Sent Events via Redis pub/sub (realtime/progress.ts)	PASS Built
Auth	HMAC-signed session cookies (custom; not JWT)	PASS Built
Hosting (web)	Vercel — SSR via @astrojs/vercel adapter	PASS Adapter wired
Hosting (API)	Railway (proposed) — Postgres + Redis + Fastify in one project	FAIL Not provisioned
Geospatial	Google Maps Geocoding + Places · curated 5-city Texas fixture as offline fallback	PASS Built (live key pending)
Email	Resend (transactional Pulse Reports)	PASS Wired (key pending)

Layer	Technology	Status
Analytics	PostHog (server-side capture planned)	WAIT Env var only — SDK not installed

3.3 Service Modules (Actual)

Implemented as modular service modules inside the Fastify runtime. **DRIFT-001** notes that BRD §12 used the word 'skills' — this implementation is service-modules rather than Claude Skill packaging. Awaiting Tim's decision; default position is to keep modules and update BRD wording.

Module	Path	Purpose
auth	apps/api/src/auth/	HMAC session cookies, origin guard, user upsert. Includes /auth/dev-login.
db	apps/api/src/db/	Drizzle schema, migrations, seed.
geospatial	apps/api/src/geospatial/	Catchment expansion (catchment.ts + googleMaps.ts) with fixture fallback.
llm	apps/api/src/llm/	5 adapters across 4 files: mock.ts, anthropic.ts, gemini.ts, openaiCompatible.ts (ChatGPT/Perplexity/Grok). Routed via index.ts.
orchestrator	apps/api/src/orchestrator/	runAudit.ts + queue.ts. BullMQ fan-out pipeline.
parsing	apps/api/src/parsing/	extractMentions.ts — mention/rank/caveat detection.
prompts	apps/api/src/prompts/	resolver.ts + inferIndustry.ts — industry → prompt set resolution.
pulse	apps/api/src/pulse/	scheduler.ts + email.ts + template.ts + unsubscribe.ts.
realtime	apps/api/src/realtime/	progress.ts — Redis pub/sub → SSE bridge.
routes	apps/api/src/routes/	audits, auth, me, admin, health, pulse — HTTP surface.
scoring	apps/api/src/scoring/	score.ts — BRD §5.2 truth table.
env / server / index	apps/api/src/	zod env validation, Fastify factory, app boot.

3.4 Data Model (Actual Schema)

Table	Key Columns	Notes
users	id, email (UNIQUE), name, oauth_provider, oauth_sub, tier (free/premium/ultra), pulse_opt_in, created_at	Free tier is the only reachable tier.

Table	Key Columns	Notes
industries	id, slug (UNIQUE), display_name, default_radius_miles	BRD \$5.4 radius lookup. 20 industries seeded.
prompt_sets	id, industry_id (FK → industries), prompts (jsonb)	Per-industry buying-intent prompts. Dental = full; 13 others share fallback.
businesses	id, user_id (FK, nullable), domain, name, address, seed_city, lat, lng, industry_id (FK), created_at	Partial unique index on (user_id, domain).
audits	id, business_id (FK), status (queued/running/complete/partial/failed), score, catchment_cities (jsonb), started_at, completed_at	One row per audit run.
audit_results	id, audit_id, llm, prompt_text, city, response_raw, mentioned, rank, has_contact_info, caveat_flag, score_band, input_tokens, output_tokens, created_at	Per-LLM × per-prompt × per-city cell. Token columns enable cost telemetry.
pulse_subscriptions	id, user_id, business_id, frequency, next_run_at, last_audit_id (FK, nullable), created_at — UNIQUE (user_id, business_id)	Drives monthly cron.
api_health	id, llm (UNIQUE), status, last_check_at, error_count	Passive — written during audit runs by createHealthBatch. No active probe.

Phase 1.5 schema additions: events table (PostHog mirror) if needed for internal analytics; Stripe-backed billing tables if premium paywall lands.

4. Milestones & Deliverables

Aligned to PLAN.md (10 milestones + 2 hardening passes). Each milestone has explicit acceptance criteria and produces a deliverable that can be demoed independently.

#	Milestone	Status	Notes
M1	Repo scaffold + CI	PASS Done	Monorepo (apps/web, apps/api, packages/shared). Lint + test.
M2	Schema + seed	PASS Done	Drizzle schema, 20 industries seeded, dental prompt set at depth.
M3	Adapter interface + mock fixtures	PASS Done	Uniform interface, deterministic mocks for all 5 providers.

#	Milestone	Status	Notes
M4	Audit orchestrator + BullMQ + SSE	PASS Done	Domain in → parallel fan-out → results in DB → SSE events.
M5	Frontend audit flow (hero → progress → reveal)	PASS Done	Including vendor comparison + CTA. Visibility score gauge + grid.
M6	Auth + user dashboard	PASS Done	HMAC session cookies. /auth/dev-login (gated, see AC-7).
M7	Pulse Report service	PASS Done	Cron + Resend email. Live-send pending key + domain.
M8	Admin dashboard (leads)	WAIT Partial	Leads UI complete. User management / health UI / funnel UI → Phase 1.5.
M9	Real-LLM adapters	WAIT Wired, gated off	LLM_USE_REAL_ADAPTERS=false. Blocked on Tim sign-off.
M10	Geospatial — Google Maps live path	WAIT Wired, key pending	Live key required. Fixture fallback already in place.
H1	Hardening — bug fixes from gate runs	PASS Done	BullMQ shared connection, SSE unsubscribe, hydration race, IPv6, pino-pretty.
H2	Hardening — deploy readiness	WAIT Partial	zod env validation PASS . Hard boot guard on dev-login → AC-7. Observability minimal.
D1	Deploy — API host provisioning	FAIL Not started	Railway provisioning, env wiring, smoke test, Vercel pointer.
D2	Deploy — production cutover	FAIL Not started	Staging → real-LLM flip → production. Gated on M9 sign-off + AC pass.

5. Acceptance Criteria

#	Criterion	Status	Verification
AC-1	Audit completes in ≤15s P95 against real LLM APIs.	FAIL Unmeasured (live)	Synthetic load, 100 audits live.
AC-2	Scoring covers all 4 cases (top-3, mentioned-not-top-3, no-contact-info, not-mentioned).	PASS Pass	12 unit tests in scoring/score.test.ts.

#	Criterion	Status	Verification
AC-3	Generic-name false-positive guard works (domain + location cross-ref).	PASS Pass	extractMentions.test.ts.
AC-4	Catchment returns sane city lists across all 20 industries.	WAIT Partial	Fixture path tested. Live path requires Google Maps key.
AC-5	Progress bar updates per-LLM in Chrome, Safari, Firefox.	WAIT Partial	Chromium PASS . Safari + Firefox manual QA → Phase 1.5.
AC-6	Signup → audit → Pulse opt-in → first email received in inbox.	WAIT Partial	Mock-mode E2E PASS . Live inbox delivery requires Resend domain.
AC-7	No dev-login endpoint reachable in production. Hard boot guard required.	FAIL Soft gate only	Currently runtime route check. Needs throw-on-boot if NODE_ENV=production and ALLOW_DEV_LOGIN unset.
AC-8	LLM API failures degrade gracefully to 'unavailable' rather than crashing audit.	WAIT Code ready	Adapter error handling PASS . Chaos test → Phase 1.5.
AC-9	Admin dashboard shows audits, lead scores, conversion funnel.	WAIT Partial	Leads PASS . Funnel + health → Phase 1.5.
AC-10	PostHog events fire on audit_started, audit_completed, vendor_clicked, consultation_booked.	FAIL Zero events	PostHog client SDK never installed. ~1h to add.

To reach Phase 1 ship-ready: AC-1 (live latency), AC-7 (boot guard), AC-10 (PostHog). AC-4 / AC-6 promote to **PASS** as soon as Google Maps and Resend keys land. AC-5 / AC-8 / AC-9 acceptable as partial for code-complete, must complete for customer-facing launch.

6. Skills, Tools, and Bots

6.1 Skills (Service Modules)

Implemented as Fastify service modules per §3.3. **DRIFT-001 open** pending Tim's decision on whether to repackage as Claude Skills in Phase 2 for cross-product agent integration (e.g., Kriss.ai).

6.2 Tools / External Integrations

Integration	Status
OpenAI API (ChatGPT) — via openaiCompatible adapter	Wired, key pending
Perplexity API — via openaiCompatible adapter	Wired, key pending
Anthropic API (Claude) — dedicated adapter	Wired, key pending
Google Gemini API — dedicated adapter	Wired, key pending
xAI API (Grok) — via openaiCompatible adapter	Wired, key pending
Google Maps Geocoding + Places	Wired, key pending
Google OAuth 2.0	Wired, credentials pending
LinkedIn OAuth 2.0	Wired, credentials pending
Resend (transactional email)	Wired, key + domain pending
Redis (BullMQ + cache + pub/sub)	Live (local)
Postgres (Drizzle)	Live (local)
PostHog (analytics)	Phase 1.5 — SDK not installed
Mapbox (geospatial fallback per SOW v1.0)	Not integrated — env-stubbed only. Decide: drop env var or wire.

6.3 Bots (Scheduled / Reactive Jobs)

- **Audit Scheduler Bot** — monthly cron, re-runs audits for Pulse Report subscribers. *Built (apps/api/src/pulse/scheduler.ts).*
- **API Health (passive)** — api_health table written during audit runs by createHealthBatch. *No active probe and no outage alerting in Phase 1.* Active probe → Phase 1.5.
- **Lead Routing Bot** — *Phase 1.5.* Phase 1 ships with /admin/leads sorted by score (no notifications, no flagging).

7. Testing Strategy

Layer	Approach	Status	Phase
Unit	Vitest. Coverage focus: parser, scorer, catchment, prompt-resolver.	PASS 67 tests passing; coverage not measured	P1

Layer	Approach	Status	Phase
Integration	Spin up Postgres + Redis. Test orchestrator with mock LLM adapters.	PASS Local; FAIL no CI pipeline yet	P1
E2E	Playwright. Hero → reveal → CTA flow.	PASS Chromium passes	P1
Load (k6 / Artillery)	100+ concurrent audits against mock + live.	FAIL Not built	P1.5
Chaos	Kill one LLM adapter mid-audit; verify graceful degradation.	FAIL Not built	P1.5
Manual QA (Safari + Firefox)	Three-browser pass on full audit flow.	FAIL Not done	P1.5

8. Deployment Plan

8.1 Environments

Environment	Purpose	State
Local	Development	PASS Running (api :4000, web :4321)
Vercel Preview	Per-PR previews. Mock adapters. Must be gated behind basic auth or allowlist until real LLMs ship.	WAIT Currently public — security artifact
Staging	Pre-prod, real LLMs at low rate, internal only.	FAIL Not provisioned
Production	Live, customer-facing.	FAIL Not provisioned

8.2 Deploy Sequence

- **Step 1 — Choose and provision API host** (Railway recommended; Fly.io / Render alternatives).
- **Step 2 — Configure env vars on host:** DATABASE_URL, REDIS_URL, SESSION_SECRET, all 5 LLM keys, OAuth credentials, Resend key, PostHog key (when SDK lands).
- **Step 3 — Deploy API; verify /health endpoint and BullMQ workers.**
- **Step 4 — Point Vercel PUBLIC_API_ORIGIN at the host URL.**
- **Step 5 — Gate Vercel preview** behind Vercel Deployment Protection or strip /auth/dev-login from preview builds. Required before any external URL share.
- **Step 6 — Add hard boot guard** on /auth/dev-login (AC-7).

- **Step 7 — Run E2E smoke against staging.**
- **Step 8 — On Tim approval of real-LLM flip + token budget:** set `LLM_USE_REAL_ADAPTERS=true` in production. Run AC-1 (live latency) and AC-8 (chaos test).

9. Risks & Mitigations

Risk	Likelihood	Impact	Mitigation
Token budget exceeds Tim's tolerance (~\$430K/yr modeled).	Medium	High	Cap city iterations. Partner-tier API access. Cache identical queries across users in same metro. Pre-flight cost telemetry already in audit_results (input/output tokens).
LLM provider TOS issues on programmatic recommendation queries.	Medium	High	Legal review before M9 flip. Identify any provider requiring partner-tier.
Mock data on public Vercel preview poisons sales trust.	High if unmitigated	High	Gate preview behind Vercel Deployment Protection. No public URL pre-launch. Mandatory before deploy step 4.
Dev-login endpoint accidentally enabled in prod.	Low	Critical	Hard boot guard: throws if <code>NODE_ENV=production</code> and <code>ALLOW_DEV_LOGIN</code> unset (AC-7).
LLM rate limiting under 1,000-rep load.	High	Medium	BullMQ throttling per provider. Distributed request pools. Partner-tier as primary mitigation. Load test in Phase 1.5.
Generic prompt fallback hurts conversion in 13 non-dental verticals.	Medium	Medium	Ship dental at depth. Verticalize others as Phase 1.5 content task.
DRIFT-001 escalates to Phase 1 rework.	Low	Medium	Default: keep services. Update BRD wording. Repackage as Skills only if/when Kriss.ai integration requires it.
Mapbox env var promises a fallback that doesn't exist.	Low	Low	Either wire Mapbox or remove <code>MAPBOX_API_KEY</code> from env. Documentation-lie risk only.

10. Exit Criteria & Handoff

Phase 1 is complete and SOW v1.1 closes when **all of the following are true**:

- AC-1 (live latency), AC-2, AC-3, AC-7 (hard boot guard) pass.

- AC-4, AC-5, AC-6, AC-8, AC-9, AC-10 either pass or are explicitly accepted as Phase 1.5 items by Tim.
- Production deploy on geotracker.iozero.ai stable for 7 consecutive days.
- Tim has approved real-LLM flip and signed off on token-budget caps.
- Sales rep playbook updated to use live URL.
- PROJECT_STATUS.md final entry written and committed.
- DRIFT-001 closed (decision recorded) or formally moved to Phase 2.

Handoff Artifacts

- Production codebase + tagged v1.0 release.
- Runbook (env vars, deploy commands, incident playbook).
- API documentation (admin endpoints, internal service interfaces).
- Post-launch monitoring dashboard (PostHog + host metrics).
- Updated BRD with any wording corrections from DRIFT resolutions.
- SOW_ALIGNMENT_v1.0.md audit and SOW v1.1 (this document).

11. Open Questions (Tim / Allan Decisions)

#	Question	Recommendation
Q1	API host: Railway, Fly.io, or Render?	Railway — single project, managed Postgres + Redis. Migrate to Fly.io later if scale demands.
Q2	DRIFT-001: keep service modules forever, or repackage as Claude Skills in Phase 2?	Keep modules. Update BRD §12 wording. Revisit only if Kriss.ai integration requires Skill dispatch.
Q3	Retrofit shadcn/ui or accept hand-styled Tailwind?	Accept hand-styled. Phase 1.5 polish pass if needed.
Q4	Dev-login lifecycle: strip on real-OAuth landing, or keep as support tool behind flag?	Strip on real-OAuth landing. Add hard boot guard now.
Q5	Mock-on-preview UX: banner, basic-auth gate, or kill public preview?	Vercel Deployment Protection (basic-auth equivalent). No public preview pre-launch.
Q6	Slow Phase 1 close-out to verticalize 13 industries, or ship and iterate?	Ship. Verticalize as Phase 1.5 content task — non-blocking.
Q7	Mapbox: drop env var or actually wire as fallback?	Drop. The 5-city fixture is the fallback. Mapbox is a Phase 2 conversation.

#	Question	Recommendation
Q8	Premium tier paywall: Phase 1.5 or defer to Phase 2?	Phase 1.5 — needed for SaaS revenue stream per BRD §9.

12. Sign-Off

Role	Name	Signature / Date
Author	Allan	_____
Approver (CEO)	Timothy Ngo	_____
Engineering Lead	(Claude Code session owner)	_____